CRUD App Infrastructure (AWS ECS + ECR + VPC)

> **Date:** 2026-05-10
> **Owner:** Xandro-D

This guide walks through creating the VPC, security groups, ECR images, ECS cluster, task definitions, services, and a bastion host for debugging.
Note: Keep the existing image URLs as-is. Suggestions for additional images are provided below each section.

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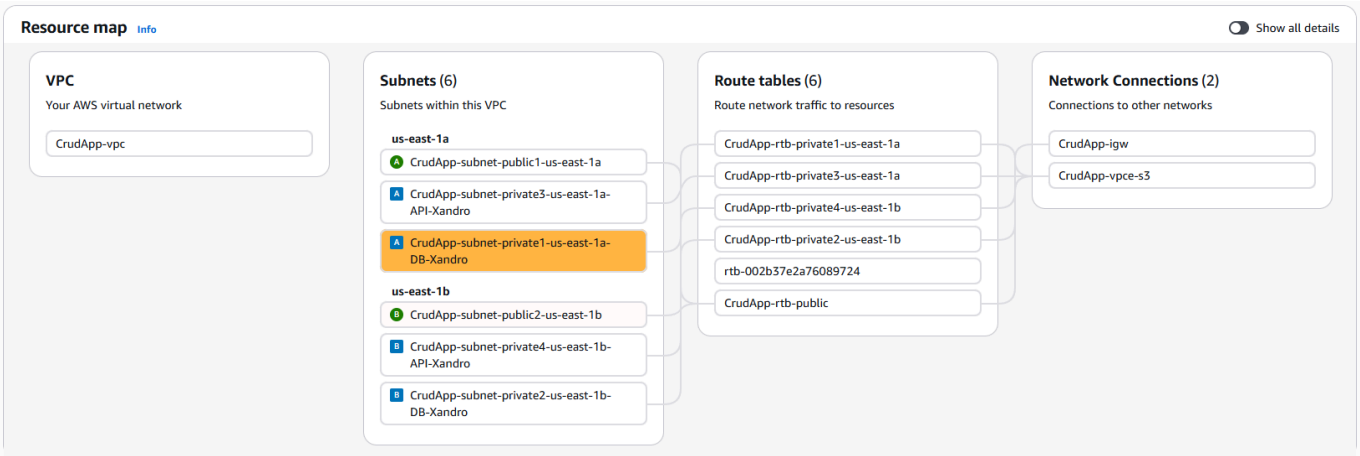
1) Create the VPC

Settings

- **Name:** `crudApp`
- **IPv4 CIDR block:** `10.0.0.0/24`
- **Availability Zones:** `2` (for better reliability)
- **Public subnets:** `2`
- **Private subnets:** `4`
  - 2 per AZ
  - 1 for API, 1 for DB per AZ
- **NAT gateways:** `1 (regional)`
- **All other settings:** default

Rename private subnets for clarity (e.g., `private-api`, `private-db`, `private-api`, `private-db`).



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2) Create Security Groups

Go to **VPC** → **Security groups** → **Create security group**.

Front-end SG

Name: `FrontEnd`
VPC: `crudlab`

Inbound rules

Type	Protocol	Port	Source	Description
HTTP	TCP	80	0.0.0.0/0	Public web access

API SG

Name: `xd-CrudApp-API`
VPC: `crudlab`

Inbound rules

Type	Protocol	Port	Source	Description
Custom TCP	TCP	8000	`FrontEnd` security group	Allow frontend to API

Bastion SG

Name: `xd-bastion-sg`
VPC: `crudlab`

Inbound rules

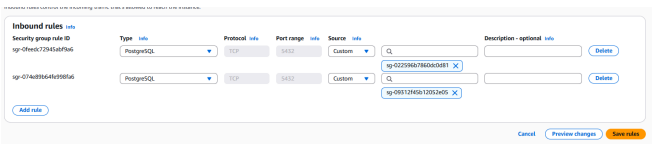
Type	Protocol	Port	Source	Description
SSH	TCP	22	0.0.0.0/0	Admin access

Database SG

Name: `xd-crudapp-db-sg`
VPC: `crudlab`

Inbound rules

Type	Protocol	Port	Source	Description
PostgreSQL	TCP	5432	`xd-CrudApp-API` security group	API access
Custom TCP	TCP	5322	`xd-bastion-sg` security group	Bastion debugging



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3) Build and Push Docker Images to ECR

Go to **ECR** → **Create repository**.

Backend image

Repository name: `xd-crudapp`

Follow the ECR push commands to build and push.

```
# API Dockerfile

FROM python:3.11-slim
COPY requirements.txt requirements.txt
RUN pip install -r requirements.txt
COPY main.py main.py
COPY ./app ./app

ENV DATABASE_URL="sqlite:///./notetaker.db" \
    SECRET_KEY="change-me" \
    ACCESS_TOKEN_EXPIRE_MINUTES="120" \
    DEBUG="False"

EXPOSE 8000

ENTRYPOINT ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000", "--
workers", "2"]
```

Frontend image

Make sure to update the frontend API path to `/api`.

```
FROM nginx
COPY ./ngnix.conf /etc/nginx/nginx.conf
COPY ./frontend ./usr/share/nginx/html

EXPOSE 80
```

Nginx config

```
worker_processes 1;

events {
    worker_connections 1024;
}

http {
    include       /etc/nginx/mime.types;
    default_type  application/octet-stream;
```

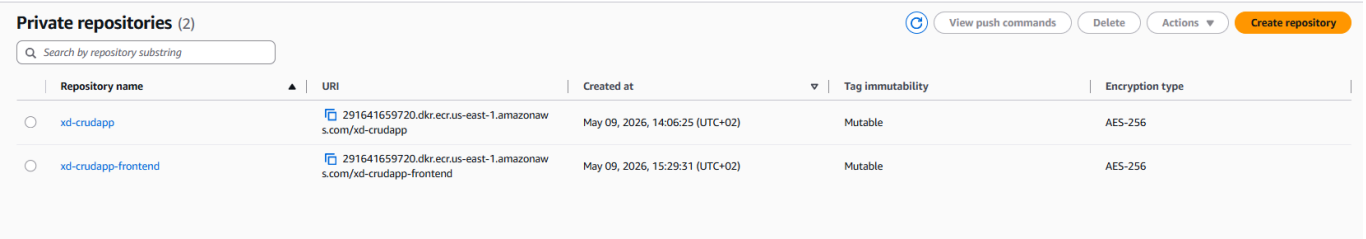
```
    sendfile        on;
    keepalive_timeout  65;

    server {
        listen 80;
        server_name _;

        # Serve frontend static files
        root /usr/share/nginx/html;
        index index.html;

        location / {
            try_files $uri $uri/ /index.html;
        }

        # Proxy API calls to private backend
        location /api/ {
            proxy_pass http://internal-BackEndLoadBallancer2-1159648751.us-east-
1.elb.amazonaws.com:8000/;
            proxy_set_header Host $host;
            proxy_set_header X-Real-IP $remote_addr;
            proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        }
    }
}
```



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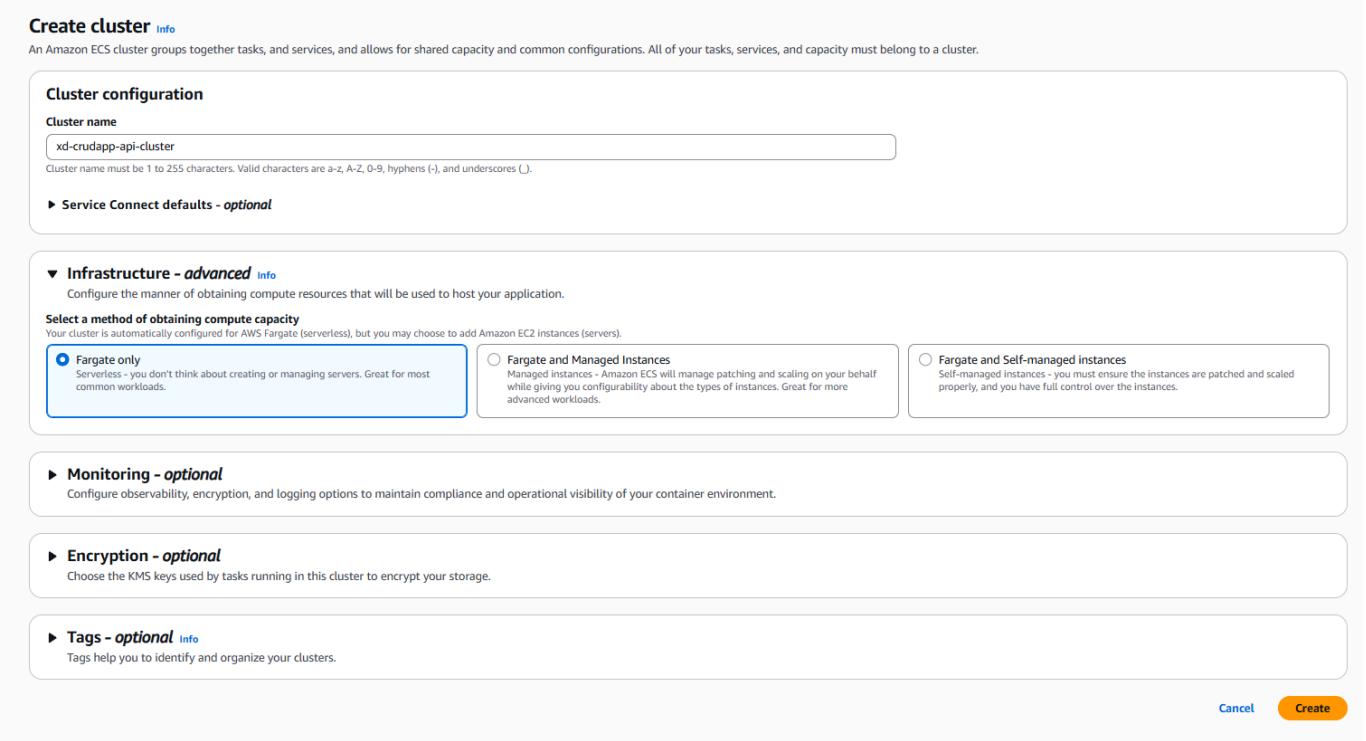
4) Create an ECS Cluster

Go to **Amazon ECS** → **Create cluster**.

- **Name:** `xd-crudapp-api-cluster`
- **All other settings:** default

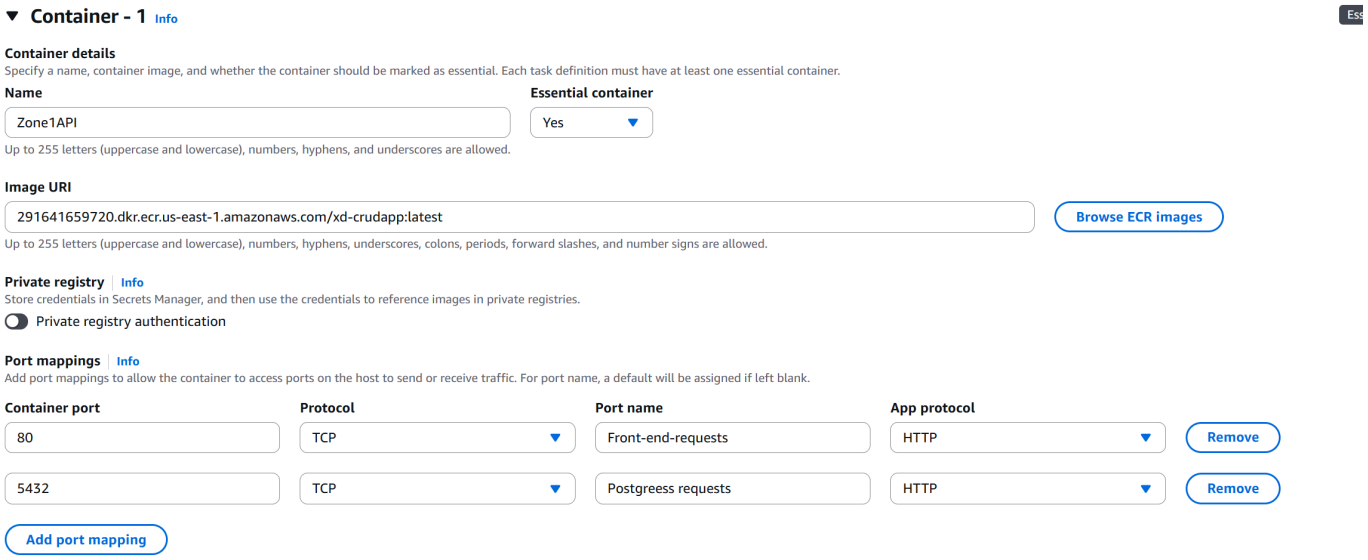


5) Create ECS Task Definitions

Go to **Amazon ECS → Task definitions → Create new**.

API Task

- **Name:** `CrudappAPI`
- **Task role:** `labrole`
- **Task execution role:** `labrole`
- **Container image:** `crudappapi`
- **Port mappings:** `8000`, `5432`



Frontend Task

- **Name:** `CrudAppFrontEnd`
- **Task role:** `labrole`

- **Task execution role:** `labrole`
- **Container image:** `crudappapi`
- **Port mappings:** `80`, `8000`

Suggested additional images:

- Task definition container config
- Port mapping screen

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6) Create Services

Frontend Service

- **Task definition:** `CrudAppFrontEnd`
- **Platform version:** latest
- **Desired tasks:** 2
- **Availability Zone rebalancing:** enabled

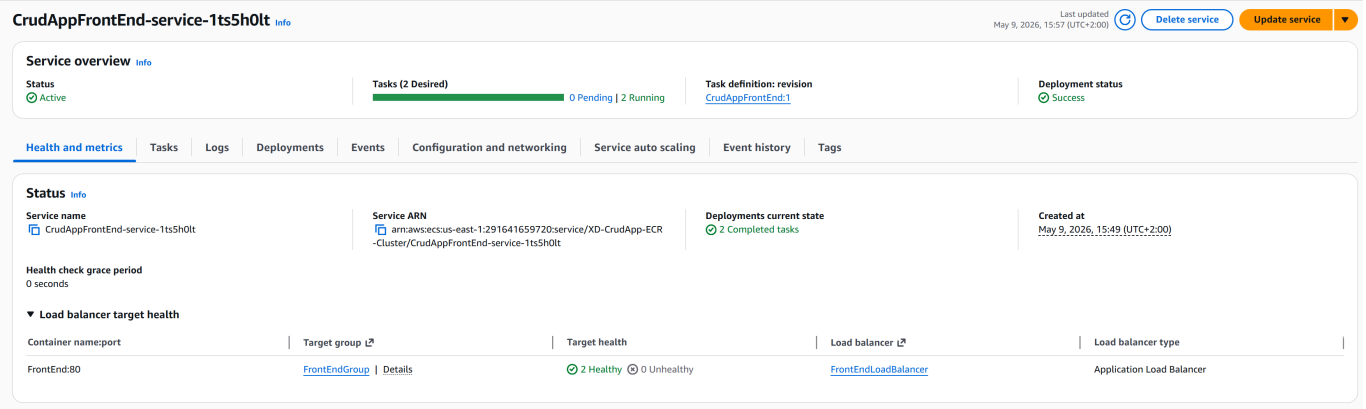
Networking

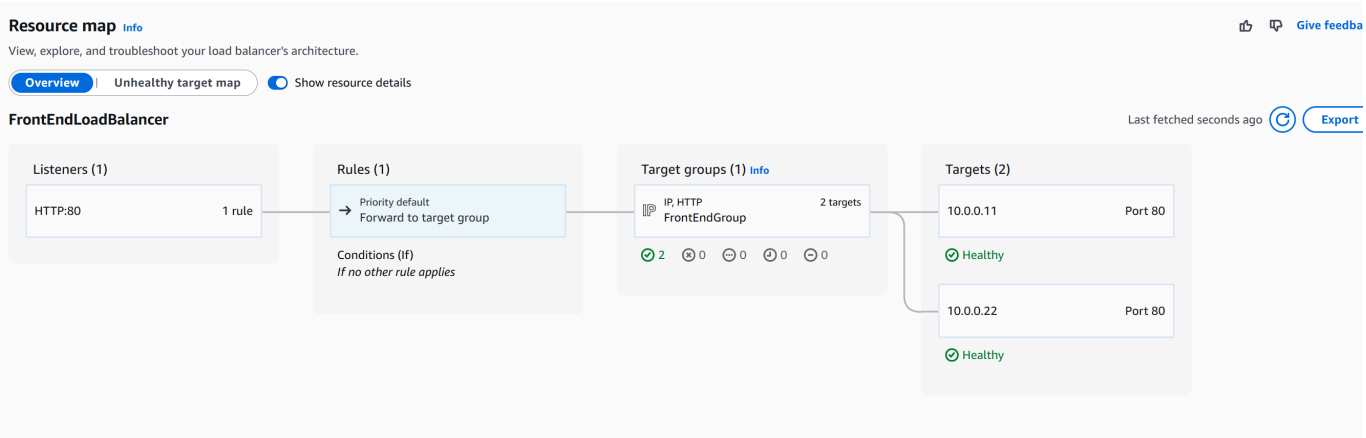
- **VPC:** `curdapp`
- **Subnets:** public subnets
- **Security group:** `FrontEnd` (create if needed)

Load balancing

- **Create new load balancer**
  - **Name:** `FrontEndLoadBalancer`
  - **Target group:** `FrontEndGroup`

Now the frontend is reachable via the load balancer.





Backend Service

Create Internal Load Balancer (before service)

Go to **EC2 → Load balancers → Create**.

- **Name:** *(your choice)*
- **Scheme:** Internal

Network mapping

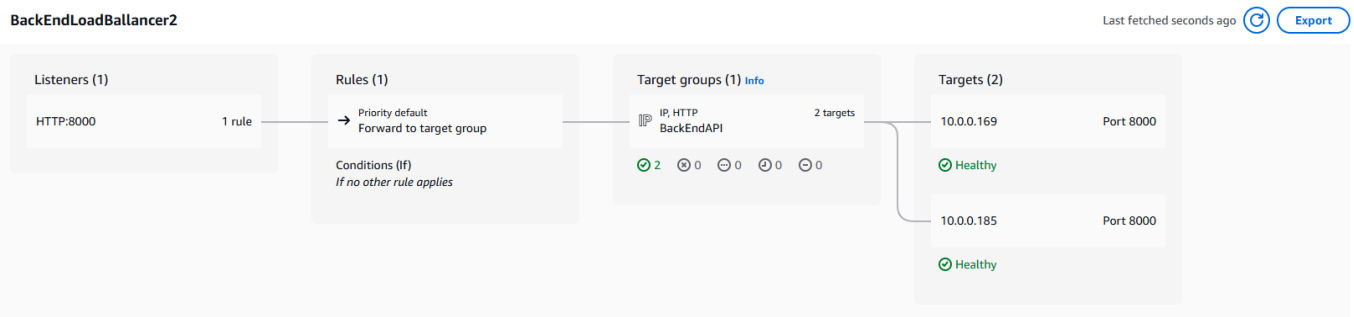
- **VPC:** `Crudapp`
- **AZ & subnets:** select the two available

Security groups

- `xd-crudapp-api`

Listeners and routing

- **Port:** 8000
- **Forward to group:** leave blank for now



Create Backend Service

- **Task definition:** `CruddAppAPI`
- **Platform version:** latest
- **Desired tasks:** 2
- **Availability Zone rebalancing:** enabled

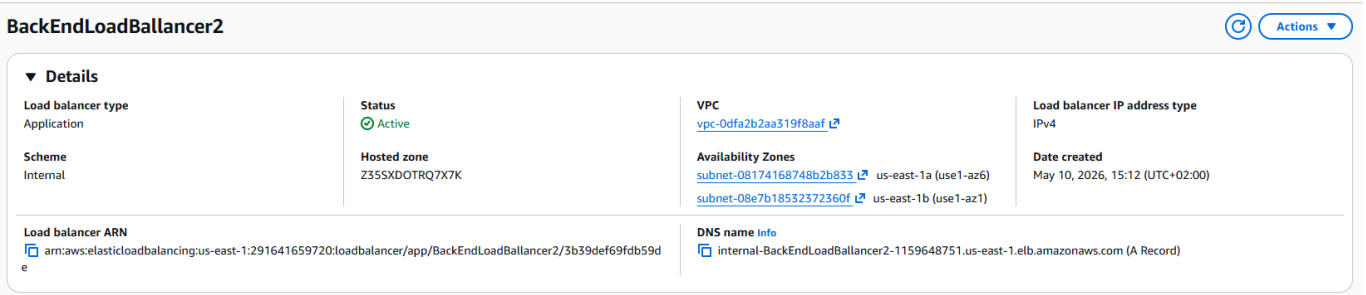
Networking

- **VPC:** `curdapp`
- **Subnets:** private API subnets
- **Security group:** `xd-CrudApp-API`
- **Public IP:** off

Load balancing

- Select existing backend load balancer

Now the backend is reachable internally via the load balancer.



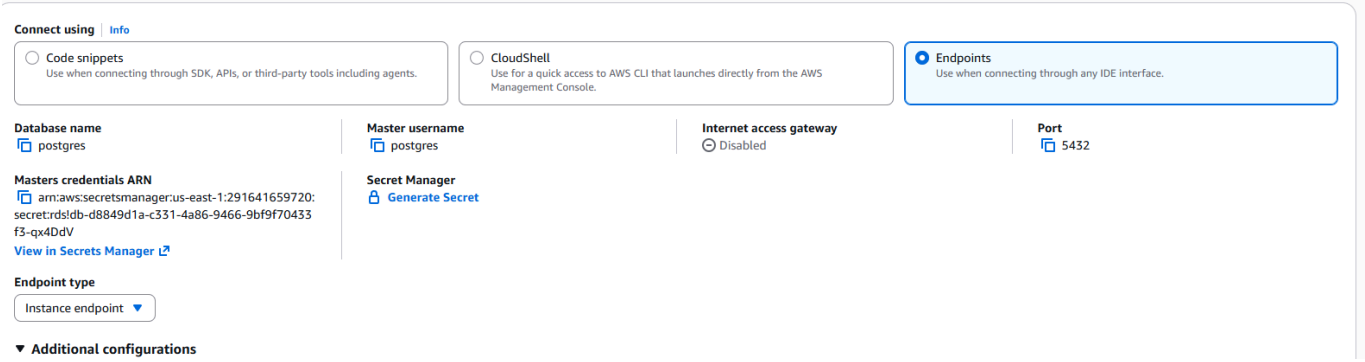
Update Nginx config to point to the internal API load balancer (already shown above).

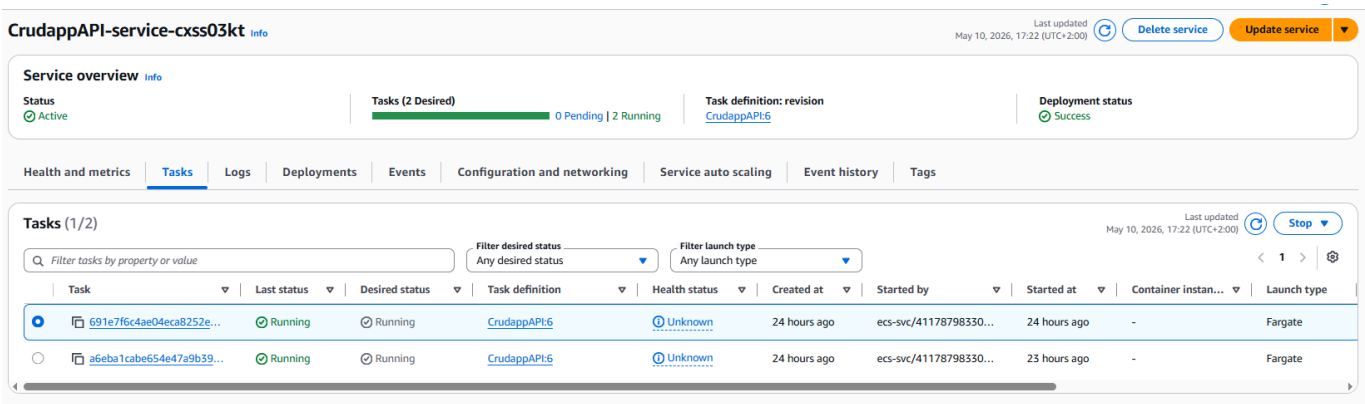
Update Backend Environment Variables

Change the backend environment variable to the RDS login string:

Location:
`Aurora and RDS → Databases → xd-curdapp-db`

> **Important:** URL-encode the password to avoid errors.





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7) Create a Bastion Host for Debugging

Go to **EC2** → **Instances** → **Launch instance**.

- **Name:** `Crudapp-Bastion`
- **SSH key:** `labkey`

AMI

Use the cheapest Ubuntu or preferred distro.

Network settings

- **VPC:** `crudapp-VPC`
- **Subnet:** public subnet 1
- **Auto-assign public IP:** enabled
- **New security group**
  - **Name:** `XD-crudapp-Bastion`
  - **Description:** `XD-crudapp-Bastion`
  - **Inbound:** SSH (default)

All other settings: default.

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8) Debugging from Bastion

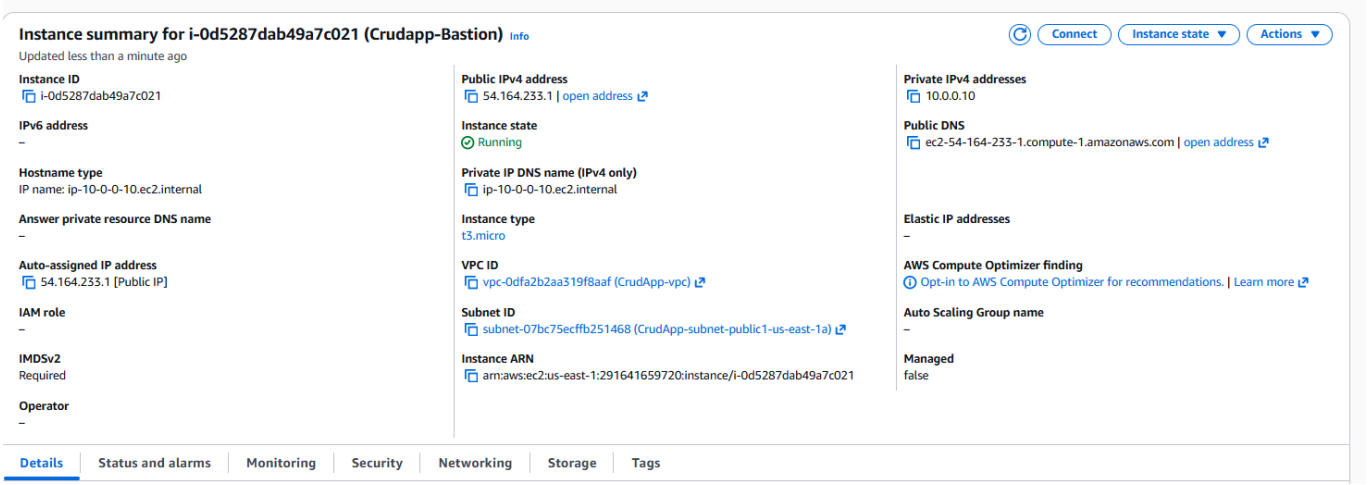
Allow DB access from bastion by updating DB security group rules. (already shown)

Then ssh into the bastion and install AWS CLI + PostgreSQL client:

```
sudo apt install awscli
sudo apt install postgresql-client
```

```
export PGPASSWORD='<database password>'
psql "host=xd-curdapp-db.czxrhpm7aefl.us-east-1.rds.amazonaws.com \
port=5432 \
dbname=postgres \
user=postgres \
sslmode=require"

export PGPASSWORD='p92Yv)IC#467dzcE1$7$LuAkNM72'
```



Changes to original code

Only two changes were made to the original code:

- Added `psycopg[binary]==3.1.18` to `requirements.txt` for PostgreSQL support.
- Updated the API base URL in the JavaScript file.